ECE 501 - DIGITAL IMAGE PROCESSING

Explainable Multi-Label Content-Based Image Retrieval for Aerial Images using Classical Computer Vision

Weekly Progress Report

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1 Introduction

During Week 5, the team started working on Phase 5 of the proposed Content-Based Image Retrieval (CBIR) system. The previous phases focused on developing a retrieval pipeline using handcrafted colour, texture, shape, and spatial features. The retrieval results and evaluation metrics obtained so far provide a foundation for identifying possible improvements to the system.

The main objective of the current phase is to move towards a more explainable and reliable retrieval system. The team began discussing how individual feature contributions can be analysed, how the fusion of different features can be improved, and how potentially unreliable retrieval results can be identified.

2 Review of Existing System

The existing system extracts handcrafted visual features from aerial images and compares the feature representations of a query image with those stored in the gallery. Different feature types capture different visual characteristics, such as colour distribution, texture patterns, object boundaries, shapes, and spatial arrangements.

The similarity scores obtained from these features can be combined to rank the gallery images according to their visual similarity to the query. The system has also been evaluated using Precision@K, Hit@K, mean Average Precision (mAP), and multi-label Precision@5.

The results obtained in the earlier phases were reviewed to understand the performance of the existing fusion approaches. This review motivated the team to investigate whether suitable feature weights, better explanations, and confidence-related analysis could improve the usefulness of the retrieved results.

3 Phase 5: Proposed Improvements

Phase 5 focuses on planning improvements to the existing retrieval framework. The following areas were identified for further investigation.

3.1 Feature-Weight Tuning

Different visual features contribute different types of information to image similarity. However, assigning equal importance to all feature cues may not always produce the best retrieval ranking.

The team started considering an approach to tune the fusion weights using a validation split. The individual feature scores can be combined using different weights, and the resulting retrieval performance can be compared to identify a suitable combination.

The aim is to determine whether a better balance between colour, texture, shape, and spatial information can improve retrieval performance. The weights will be investigated systematically rather than selected only by assumption.

3.2 Explainability of Retrieval Results

Another objective is to make the retrieval process easier to understand. Instead of displaying only the final similarity score, the system is planned to provide a breakdown of the contributions made by individual feature cues.

For example, the explanation could indicate whether a retrieved image has a high similarity because of its colour distribution, texture, shape, or spatial characteristics. Region-based visualizations or heat maps will also be explored to highlight image areas that contribute to similarity.

These explanations are intended to help users understand why particular aerial images appear among the top retrieved results.

3.3 Uncertainty and Reliability Analysis

A high-ranked result is not necessarily always reliable. Therefore, the team started discussing methods to identify retrievals for which the system may be uncertain.

One proposed approach is to examine the difference between the similarity scores of the top-ranked candidates. A small difference between competing candidates may indicate that the system has difficulty distinguishing between them.

Another possibility is to analyse disagreement between individual feature cues. For example, colour similarity may be high while shape similarity is low. Such disagreement could be used as an indicator for further investigation.

These approaches will be explored to flag potentially unreliable results without introducing machine learning or deep learning methods.

4 Evaluation Plan

The planned improvements will be evaluated using held-out query images so that performance can be assessed on queries not used for tuning the feature weights.

Precision@K will measure the proportion of relevant images among the top-ranked results, while mAP will measure the quality of the overall retrieval ranking. Multi-label relevance will also be examined using Jaccard similarity thresholds ranging from 0.5 to 0.8.

The results obtained with the proposed improvements will be compared with the existing retrieval approach. This comparison will help determine whether the changes improve retrieval quality and whether the additional explanations provide useful information about the system's decisions.

5 Work Completed

During Week 5, the following work was carried out:

- Initiated work on Phase 5 of the CBIR project.
- Reviewed the existing feature extraction, similarity computation, and fusion pipeline.
- Examined the earlier retrieval results to identify areas for possible improvement.
- Discussed validation-based tuning of individual feature-fusion weights.
- Planned a feature-wise score breakdown and region-based visualizations for explainability.
- Explored the use of similarity-score margins and feature-cue disagreement to identify potentially unreliable retrievals.
- Defined a preliminary evaluation plan using Precision@K, mAP, and multi-label Jaccard similarity thresholds.

6 Plan for Next Week

The next stage will focus on developing and testing the ideas planned during this week.

- Begin implementing the selected Phase 5 components.
- Investigate feature-weight tuning on the validation split.
- Develop a method to display individual feature contributions to the final similarity score.
- Explore region-based heat maps for explaining retrieval results.
- Implement and test indicators of uncertainty based on score margins and feature disagreement.
- Evaluate the updated system on held-out queries and compare its performance with the existing approach.

7 Conclusion

Week 5 marked the beginning of Phase 5, with the team reviewing the existing CBIR pipeline and planning improvements to its retrieval process. The main areas identified were feature-weight tuning, explainability, uncertainty analysis, and systematic evaluation. These improvements aim to make the retrieval results more interpretable and reliable while maintaining the use of classical computer vision and handcrafted features. The next stage will focus on implementing and evaluating the proposed components.